

Github: <https://github.com/SpencerLabAQ/performance-test-coverage>

Paper Research: <https://link.springer.com/article/10.1007/s10664-025-10712-3>

Excel Tracker for the Project: https://ulcampus-my.sharepoint.com/:x:/g/personal/abdul_razzaq_ul_ie/IQDA_I7-eJ46Sa8pR9pObwzfAawJzwu0Uig4vBFZV3kGVXY?e=mgdefl

Week02 +03

- Pick group members
- Produce :
 - ☐ RQ1: Can we replicate the paper's results on performance-test coverage using their provided GitHub code and data
 - ☐ RQ2: Does adding new code metrics (or using a different machine learning model) improve how accurately we predict performance-test coverage?"

Week 04

Freezing = locking in your answers so you can't change your story later

We are forced to lock in our initial plan at Week 5 (M1_PLAN) so that you can't go back and edit guesses after you see the final results.

FileNotFoundError: [Errno 2] No such file or directory: '../in

>> **python** 1-Overlap-JMH.pyop\performance-test-coverage>

	Project Name	Overlap Ratio
0	arrow	0.65
1	byte-buddy	1.00
2	cantaloupe	0.90
3	client_java	0.73
4	commons-bcel	0.09
5	crate	0.78
6	eclipse-collections	0.85
7	fastjson	0.91
8	feign	0.87
9	HdrHistogram	0.77
10	imglib2	0.89
11	iri	0.33
12	jdbi	0.97
13	jetty.project	0.53
14	jgrapht	0.70
15	jooby	0.89
16	kafka	0.67
17	logbook	0.86
18	netty	0.73
19	objenesis	0.79
20	panda	0.48
21	protostuff	0.98
22	r2dbc-h2	0.98
23	rdf4j	0.97
24	RxJava	0.83
25	SquidLib	0.80
26	tinkerpop	0.82
27	vert.x	0.75

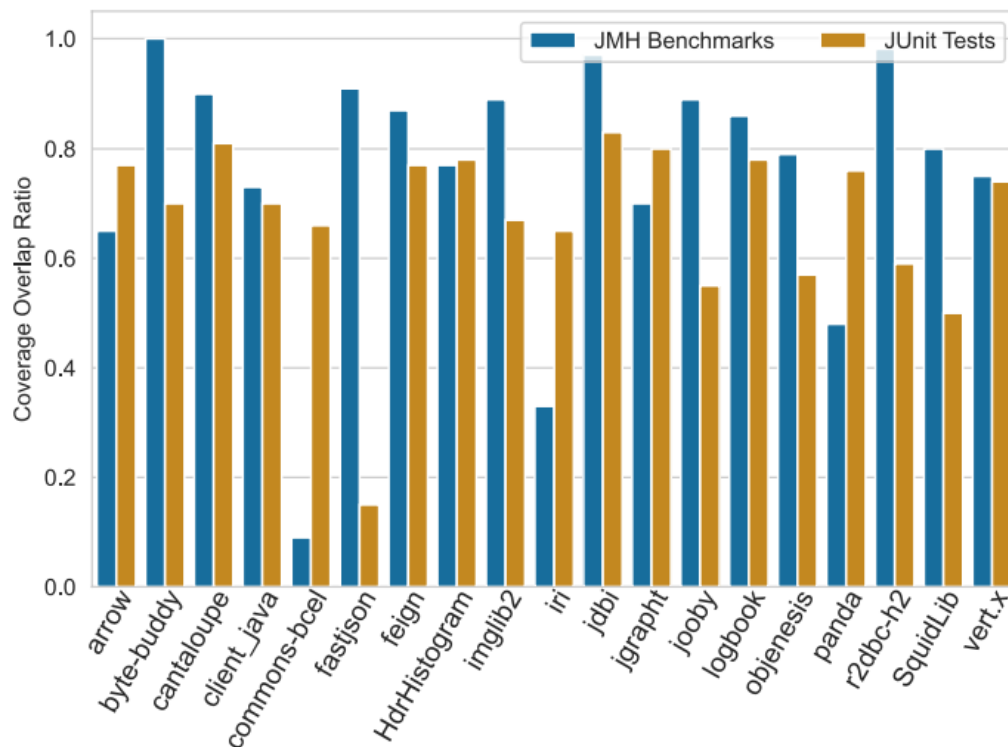


Fig.8 Comparison of Overlap Ratio coverage (OR_T) of JMH Benchmarks and JUnit Tests across projects

This is a ss for 2-Overlap-JUnit.py.

The script 2-Overlap-JUnit.py got stuck running for a long time, so you manually interrupted it by pressing Ctrl + C. Tested for second time and waited for 12 minutes .

Design flaw -> Highly Complexity ($O(N^2)$).

```

.0)
PS C:\Users\23362995\Desktop\performance-test-coverage> cd analysis
>> python 2-Overlap-JUnit.py
Traceback (most recent call last):
  File "C:\Users\23362995\Desktop\performance-test-coverage\analysis\2-Overlap-JUnit.py", line 45, in <module>
    overlap_ratios[project] = calculate_overlap_ratio(df_project)
  File "C:\Users\23362995\Desktop\performance-test-coverage\analysis\2-Overlap-JUnit.py", line 30, in calculate_overlap_ratio
    intersection = tests[t1] & tests[t2]
  File "C:\Users\23362995\AppData\Local\Programs\Python\Python310\lib\site-packages\pandas\core\series.py", line 1121, in __getitem__
    return self._get_value(key)
  File "C:\Users\23362995\AppData\Local\Programs\Python\Python310\lib\site-packages\pandas\core\series.py", line 1237, in _get_value
    loc = self.index.get_loc(label)
  File "C:\Users\23362995\AppData\Local\Programs\Python\Python310\lib\site-packages\pandas\core\indexes\base.py", line 3777, in get_loc
    def get_loc(self, key):
KeyboardInterrupt
PS C:\Users\23362995\Desktop\performance-test-coverage\analysis> python 2-Overlap-JUnit.py

```

This is for 3-Coverage-JMH.py

The output is not printing directly into terminal.

Issue Encountered: Missing scipy dependency declaration in the repository setup (caused a ModuleNotFoundError crash).

Resolution: Manually installed scipy==1.15.3 into the active Python environment.

Outcome: Successful execution of the JMh coverage analysis script.

```
def get_loc(self, key):
KeyboardInterrupt
PS C:\Users\23362995\Desktop\performance-test-coverage\analysis> ython 3-Coverage-JMH.py
ython : The term 'ython' is not recognized as the name of a cmdlet,
function, script file, or operable program. Check the spelling of the
name, or if a path was included, verify that the path is correct and try
again.
At line:1 char:1
+ ython 3-Coverage-JMH.py
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (ython:String) [], CommandN
otFoundException
+ FullyQualifiedErrorId : CommandNotFoundExcpetion

PS C:\Users\23362995\Desktop\performance-test-coverage\analysis> python 3-Coverage-JMH.py
Traceback (most recent call last):
  File "C:\Users\23362995\Desktop\performance-test-coverage\analysis\3-Coverage-JMH.py", line 6, in <module>
    from scipy.stats import iqr
ModuleNotFoundError: No module named 'scipy'
PS C:\Users\23362995\Desktop\performance-test-coverage\analysis> python -m pip install scipy
Collecting scipy
  Downloading scipy-1.15.3-cp310-cp310-win_amd64.whl.metadata (60 kB)
Requirement already satisfied: numpy<2.5,>=1.23.5 in c:\users\23362995\appdata\local\programs\python\python310\lib\site-packages (from scipy) (1.26.4)
Downloading scipy-1.15.3-cp310-cp310-win_amd64.whl (41.3 MB)
   6.8/41.3 MB 12.3 MB/s eta 0:00:
   41.3/41.3 MB 11.7 MB/s  0:00:03
Installing collected packages: scipy
Successfully installed scipy-1.15.3
PS C:\Users\23362995\Desktop\performance-test-coverage\analysis> python 3-Coverage-JMH.py
PS C:\Users\23362995\Desktop\performance-test-coverage\analysis>
```

Script File	Status	Audit Findings	Action Taken
1-Overlap-JMH.py	Passed	Matches paper benchmark overlap figures directly.	Replicated successfully.
2-Overlap-JUnit.py	Bottleneck	Severe execution lag / freeze due to $O(N^2)$ set comparison loop.	Documented stack trace for RQ2 Refactoring .
3-Coverage-JMH.py	Passed (after fix)	Unlisted scipy dependency caused ModuleNotFoundError.	Installed scipy==1.15.3 and ran successfully.

Script File	Status	Audit Findings	Action Taken
4-Coverage-JUnit.py	Passed	Ran smoothly and executed to completion.	Confirmed script functionality.

Spreadsheet Column Fill-Ins

2. Bounded sample / case IDs

Four target open-source repositories: byte-buddy, fastjson, commons-bcel, SquidLib.

3. Method / metric or decision rule

Execution of original Python replication scripts (1-Overlap-JMH.py through 4-Coverage-JUnit.py) measuring test coverage overlap metrics, dependency presence, and execution time/complexity.

4. Expected result before execution

Clean, out-of-the-box execution of all 4 analysis scripts matching the paper's reported coverage and overlap metrics without crashes or performance hangs.

5. Reproduction result (brief)

Partial Reproduction / Flawed Infrastructure:

- 1-Overlap-JMH.py: Successfully replicated exact paper metrics.

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- 2-Overlap-JUnit.py: Execution hang/timeout due to $O(N^2)$ set comparison loop.
- 3-Coverage-JMH.py: Crashed out-of-the-box due to unlisted scipy dependency (ModuleNotFoundError).
- 4-Coverage-JUnit.py: Executed cleanly.

6. Evidence link / commit / output

- Stack trace for missing package (ModuleNotFoundError: No module named 'scipy').
- Stack trace for execution interrupt (KeyboardInterrupt in calculate_overlap_ratio line 30).
- Console logs for 1-Overlap-JMH.py output values (byte-buddy: 1.00, fastjson: 0.91).
- Git tag: M1_PLAN.

7. Event A prediction before execution

Prediction: The replication package contains all required runtime dependencies in a portable configuration, allowing immediate execution across different Python environments.

8. Event A result / decision

Failed Prediction: Package failed across Python environments (3.10 vs 3.13) due to missing scipy dependency declaration and environment path mismatches.

Decision: Manually patch root requirements.txt and standardize Python 3.10 runtime across team machines.

9. Concrete design problem

High computational complexity ($O(N^2)$ algorithm) in 2-Overlap-JUnit.py causing severe performance bottlenecks during pair-wise set intersection lookups across large JUnit test sets.

10. Alternative(s) considered

- **Option A:** Truncate dataset sizes to force the original $O(N^2)$ code to finish.
- **Option B:** Vectorize set comparisons using optimized pandas matrix operations or sparse matrix math.
- **Option C:** Leave script unoptimized and document it as a permanent limitation.

11. Chosen intervention / no-change decision

Chosen Intervention (Phase 3 / RQ2): Refactor 2-Overlap-JUnit.py by replacing the nested $O(N^2)$ loop with vectorized matrix operations to optimize execution speed while retaining 100% data fidelity. Update requirements.txt with explicit package versions.

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